

## Numbers CW

1. Find the largest four-digit number exactly divisible by 55.  
(a) 9900 (b) 9955 (c) 9945 (d) 9960
2. What must be added to 581173 to make it exactly divisible by 11?  
(a) 1 (b) 2 (c) 0 (d) 3
3. What least number must be added to 15763 so that it is exactly divisible by 18?  
(a) 5 (b) 18 (c) 12 (d) 13
4. What least number must be subtracted from 178669 so that it is exactly divisible by 36?  
(a) 1 (b) 18 (c) 5 (d) 36
5. Find the LCM of  $\frac{4}{3}$ ,  $\frac{8}{9}$ ,  $\frac{3}{5}$   
(a) 20 (b) 24 (c)  $\frac{1}{24}$  (d)  $\frac{1}{20}$
6. Find the HCF of  $\frac{2}{5}$ ,  $\frac{12}{11}$ ,  $\frac{1}{3}$   
(a)  $\frac{1}{165}$  (b)  $\frac{5}{165}$  (c)  $\frac{2}{165}$  (d)  $\frac{12}{165}$
7. Find the LCM and HCF of 0.25, 0.5, 0.75  
(a) 7.5, 0.05 (b) 0.75, 0.5 (c) 0.75, 0.05 (d) 7.5, 0.5
8. The product of 2 numbers is 3024 and their LCM is 36. Find their HCF.  
(a) 88 (b) 82 (c) 84 (d) 86
9. The LCM of two numbers is 36 times that of their HCF. The product of numbers is 3600. Find their HCF,  
(a) 10 (b) 15 (c) 8 (d) 20
10. LCM of two numbers is 12 times of their HCF. Sum of LCM and HCF is 195. If one of them is 60. Find the other.  
(a) 48 (b) 45 (c) 52 (d) 36
11. The LCM of two numbers 45 times their HCF. If one number is 250 and the sum of their HCF and LCM is 2300. Find the other number.  
(a) 400 (b) 450 (c) 460 (d) 420
12. The sum of two numbers is 135 and their HCF is 9. How many such pairs of numbers can be formed?  
(a) 6 (b) 2 (c) 5 (d) 4
13. HCF of two numbers is 12 and their product is 3600. How many such pairs of numbers can be formed?  
(a) 0 (b) 1 (c) 2 (d) 4
14. Find the smallest number which when decreased by 4 is exactly divisible by 9, 12, 15 and 18.  
(a) 188 (b) 182 (c) 186 (d) 184
15. Find the least number which when divided by 12, 15, 20 and 24 leaves the same remainder 2 in each case.  
(a) 122 (b) 120 (c) 124 (d) 126
16. Find the least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 but when divided by 9 leaves no remainder.  
(a) 1683 (b) 1725 (c) 1685 (d) 1723

17. Five bells begin to toll together and toll at intervals of 24, 40, 64, 72 and 120 s. After what interval of time will they toll again together?  
 (a) 42 min (b) 36 min (c) 48 min (d) 54 min
18. Three athletes start together to travel the same way around on a circular path of 12 km with speeds of 3 km/ph, 4 km/ph and 6 km/ph respectively. They meet at the same point again after what time?  
 (a) 12 h (b) 8 h (c) 16 h (d) 10 h
19. The traffic lights at three different road crossings change after every 48, 72 and 108 s respectively. If they change simultaneously at 9 am, at what time will they change again simultaneously?  
 (a) 10 min 10 s (b) 7 min 12 s (c) 7 min 10 s (d) 12 min 10 s
20. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?  
 (a) 4 (b) 10 (c) 15 (d) 16
21. A man has 324 red balls, 396 white balls and 428 green balls. He wants to arrange them into groups of equal size without mixing. Find the least number of groups required to arrange them.  
 (a) 285 (b) 287 (c) 284 (d) 289
22. The length, breadth and height of a room are 6 m 30 cm, 5 m 85 cm and 3 m 60 cm respectively. What will be the greatest length of a tape which can measure the dimensions of room exact number of times?  
 (a) 45 cm (b) 50 cm (c) 60 cm (d) 40 cm
23. Find the unit digit of the expression  $73^{54}$ .  
 (a) 1 (b) 9 (c) 7 (d) 3
24. Find the unit digit of the expression  $55^{725} + 73^{5810} + 22^{853}$ .  
 (a) 4 (b) 0 (c) 6 (d) 5
25. The unit digit of the expression is  $125^{813} \times 553^{3703} \times 4532^{828}$   
 (a) 4 (b) 2 (c) 0 (d) 5
26. Find the number of zeroes at the end of 50!  
 (a) 13 (b) 11 (c) 5 (d) 12
27. Find the number of zeroes at the end of 1400!  
 (a) 347 (b) 336 (c) 280 (d) 349
28. Three pens, four pencils and five copies cost Rs. 750. Ten copies, six pens and 9 pencils cost Rs. 1580. What is the cost of six pens, ten copies and 4 pencils?  
 (a) 1280 (b) 1180 (c) 1080 (d) Cannot be determined
29. The cost of four pens, six pencils and sixteen copies is Rs. 500, while the cost of seven pens, nine pencils and nineteen copies is Rs. 620. What is the cost of one pen, one pencil and one copy?  
 (a) 120 (b) 40 (c) 150 (d) Cannot be determined
30. Ajay sold 38 pieces of clothing (combined in the form of shirts, trousers and ties. If he sold at least 11 pieces of each item and he sold more shirts than trousers and more trousers than ties, then the number of ties that he must have sold is:  
 (a) Exactly 11 (b) At least 11 (c) At least 12 (d) Cannot be determined

31. Abhay is sitting on vertex P of a square PQRS. It starts jumping to the immediately adjacent vertex on either side in random fashion and stops when it reaches point R. In how many ways can it reach point R if it makes exactly 7 jumps?  
(a) 1 (b) 3 (c) 5 (d) 0
32. Arjun earns Rs. 40 the first day and spends Rs. 25 on the second day. He earns Rs. 40 on the third day and spends Rs. 25 on the fourth day and so on. On which day would he have Rs. 100?  
(a) 10 (b) 9 (c) 8 (d) 11
33. Ajay is typing numbers 5 to 700 on his computer. How many times does he press the key of his computer?  
(a) 988 (b) 1988 (c) 1888 (d) 1990
34. Ankit is typing numbers from 50 to 1500 on a computer. How many times does the person press the keys?  
(a) 4,804 (b) 4,750 (c) 4,600 (d) 5,000
35. A five-digit number is taken. Sum of the first four digits (excluding the number at the unit's digit) equals sum of all the five digits. Which of the following will not divide this number necessarily?  
(a) 10 (b) 2 (c) 4 (d) 5
36. How many factors are for 6000?  
(a) 45 (b) 60 (c) 40 (d) 30
37. How many factors are for 4200?  
(a) 40 (b) 48 (c) 42 (d) 60
38. How many factors are for 3200?  
(a) 48 (b) 40 (c) 36 (d) 32
39. How many factors are for 4900?  
(a) 27 (b) 30 (c) 49 (d) 36
40. How many factors are for 1080?  
(a) 30 (b) 48 (c) 36 (d) 27

## Numbers HW

1. What is the smallest-number by which 1250 be divided in order to make it a perfect square?  
(a) 5 (b) 25 (c) 2 (d) 4
2. A boy had to do a multiplication. Instead of taking 35 as one of the multipliers, he took 53. As a result, the product went up by 540. What is the new product?  
(a) 1590 (b) 1450 (c) 1550 (d) 1420
3. The number of Students in three sections of a medical College are 40, 45 and 60. Find the minimum number of Books required for the class library for equal distribution in the three sections?  
(a) 340 (b) 360 (c) 320 (d) 380
4. A worker was engaged for a certain number of days and was promised to be paid Rs.1755. He remained absent le: some days and was paid Rs.1365 only. What were his daily wages?  
(a) Rs.182 (b) Rs.195 (c) Rs.185 (d) Rs.192
5. A heap of stones can be made up into groups of 7. When made up into groups of 16, 18, 20 and 25 there are 4 stones left in each case. How many stones at least can there be in the heap?  
(a) 18000 (b) 18004 (c) 18002 (d) 18006
6. A sum of Rs.640 is distributed among three persons Pravin, Ankus and Ram. It Ankus gets Rs.70 more than Pravin. R Ram gets Rs.80 more than Ankus. Find the share of Ankus.  
(a) Rs.210 (b) Rs.215 (c) Rs.200 (d) Rs.220
7. There were 240 multiple choice questions in an entrance examination of a management college. A candidate was given 2 marks for every correct answer and penalized  $\frac{1}{2}$  mark for every wrong answer. The candidate scored 160 marks in the paper having answered all the questions. Find the number of correct answers given by the candidate?  
(a) 108 (b) 112 (c) 116 (d) 104
8. Find the least number which when increased by 3 is exactly by 10, 12, 14 and 16?  
(a) 1680 (b) 1677 (c) 1697 (d) 1670
9. Find the least number of girls so that they can be arranged in the groups of 15 or 20 or 30?  
(a) 60 (b) 90 (c) 45 (d) 80
10. Find the least perfect square number which is exactly divisible by 3, 4, 5, 6 and 8?  
(a) 3600 (b) 3900 (c) 3500 (d) 3200
11. Find the least number which when divided by 12, 15 and 20 leaves remainders 4, 7 and 12 respectively?  
(a) 48 (b) 60 (c) 52 (d) 64
12. Out of four consecutive prime numbers, the product of first three is 385 and the product of the last three is 1001. Find the last number?  
(a) 7 (b) 11 (c) 13 (d) 17
13. Find the smallest number to be added to the least three-digit number so that it is exactly divisible by 45?  
(a) 35 (b) 25 (c) 45 (d) 15
14. Find the sum of the smallest four-digit number and the largest five-digit number?  
(a) 199900 (b) 100999 (c) 211000 (d) 211999

15. Find the unit digit in the expansion of  $42^{23}$ .

- (a) 4 (b) 8 (c) 6 (d) 2

16. Find the unit digit in the expansion of  $325^{568}$  is

- (a) 0 (b) 5 (c) 1 (d) 2

17. Find the unit digit in the expansion of  $314^{98}$

- (a) 6 (b) 4 (c) 8 (d) 2

18. The traffic lights at three different road crossings change after every 50 s, 75 s and 100 s respectively. If they start changing simultaneously at 10 am, after how much time will they change again simultaneously?

- (a) 6 min (b) 5 min (c) 3 min (d) 8 min

19. Ajay and Bablu had some candies each. If Ajay gave 5 candies to Bablu, then Bablu will have twice as many as Ajay. Instead of that, if Bablu were to give 5 candies to Ajay, then they will both have the same number of candies. How many did Ajay have?

- (a) 28 (b) 21 (c) 26 (d) 25

20. Find the least number which when divided by 15, 20 and 24 leaves 4 as remainder but when divided by 13 leaves no remainder.

- (a) 364 (b) 370 (c) 484 (d) 244

## Answer Key

|       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. b  | 2. b  | 3. b  | 4. d  | 5. d  | 6. d  | 7. a  | 8. a  | 9. b  | 10. c |
| 11. b | 12. d | 13. c | 14. b | 15. c | 16. d | 17. b | 18. b | 19. d | 20. a |